

Assessment_2_report

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1. Introduction

This report documents the importing, preliminary exploration, merging and preparation, cleaning and variable re-coding, and filtering and reshaping of NHANES demographic and blood-pressure datasets from the 2013–2014, 2015–2016, and 2017–2018 survey cycles.

For each section, the report first explains the steps taken, then shows the code, then shows a snippet of the key output (tables are trimmed to the most informative rows), and finally interprets the result. All figures quoted in the text are calculated directly from the data, so they update automatically each time the report is knitted.

2. Data Import and Preliminary Exploration

This section imports the six NHANES files (a demographic file and a blood-pressure examination file for each of the three cycles) and inspects their dimensions and structure before any changes are made. Getting to know the raw data first shows which variables need cleaning or re-coding later.

Importing the Datasets

The `import()` function from the `rio` package reads each SAS transport (`.xpt`) file, and `here()` builds the file path from the project folder so that the report runs on any computer.

Data Dimensions

The dimensions show the number of rows (observations) and columns (variables) in each dataset. They were obtained with `dim()` and collected into one table so the six files can be compared at a glance.

Table 1: Dimensions of the raw NHANES datasets

cycle	dataset	rows	columns
2013-2014	Demographic	10175	47
2013-2014	Blood pressure	9813	23
2015-2016	Demographic	9971	47
2015-2016	Blood pressure	9544	21
2017-2018	Demographic	9254	46
2017-2018	Blood pressure	8704	21

The demographic datasets ranged from 9,254 rows and 46 columns in 2017–2018 to 10,175 rows and 47 columns in 2013–2014.

The blood pressure datasets ranged from 8,704 rows and 21 columns in 2017–2018 to 9,813 rows and 23 columns in 2013–2014.

The number of columns differed slightly between cycles because NHANES adds and removes some variables from one cycle to the next, which is why later steps select only the columns that are common to the analysis.

Variable Structures

The structure of each dataset was examined with `str()` to review variable names and data types, and `summary()` was used to see the range and typical values of key variables. A snippet of the first rows of the variables of interest is displayed below.

Table 2: Snippet of the raw demographic data (2013-2014)

SEQN	RIAGENDR	RIDAGEYR	RIDRETH1	DMDEDUC2	DMDMARTL
73557	1	69	4	3	4
73558	1	54	3	3	1
73559	1	72	3	4	1
73560	1	9	3	NA	NA
73561	2	73	3	5	1
73562	1	56	1	4	3

Table 3: Snippet of the raw blood pressure data (2013-2014)

SEQN	BPXSY1	BPXDI1	BPXSY2	BPXDI2	PEASCCT1
73557	122	72	114	76	NA
73558	156	62	160	80	NA
73559	140	90	140	76	NA
73560	108	38	102	34	NA
73561	136	86	134	88	NA
73562	160	84	158	82	NA

The demographic datasets include identifiers, quantitative measurements, categorical variables stored numerically. SEQN identifies respondents, while RIDAGEYR represents age. Variables such as RIAGENDR, RIDRETH1, DMDEDUC2, and DMDMARTL are categorical, although stored numerically.

The blood pressure datasets also include identifiers, measurements, and categorical variables stored numerically. BPXSY1 and BPXDI1 represent quantitative systolic and diastolic blood pressure measurements, while BPARM, BPCSZ, and BPXPTY represent coded categories.

Thus, some numeric variables, such as RIAGENDR, RIDRETH1, BPARM, and BPCSZ, would be more appropriately treated as factors for categorical analysis. SEQN serves as an identifier, while age and blood pressure measurements are appropriately stored as numeric values.

3 Datasets Merging and Preparation

This section combines the demographic and blood pressure datasets across the three survey cycles and keeps only the columns needed for the analysis.

Column Selection

The columns to keep were listed in two vectors, `pressure_keep` and `demographic_keep`. The `select()` function with `any_of()` keeps only those columns that exist in a given cycle, so a variable missing from one cycle does not cause an error. The `mutate()` function was then used to create `wave_id`, which records the survey cycle each row came from.

Combine Datasets

The `bind_rows()` function stacked the selected columns from the three cycles, one combined blood pressure dataset (`pressure_all_cycles`) and one combined demographic dataset (`demographic_all_cycles`). Because the rows are stacked together, `wave_id` is what keeps track of the cycle for each record.

Table 4: Blood pressure records per survey cycle

wave_id	n
2013-2014	9813
2015-2016	9544
2017-2018	8704

Table 5: Demographic records per survey cycle

wave_id	n
2013-2014	10175
2015-2016	9971
2017-2018	9254

The combined blood pressure dataset, `pressure_all_cycles`, contains 28,061 rows and 11 columns, while the combined demographic dataset, `demographic_all_cycles`, contains 29,400 rows and 10 columns.

Duplicate Check on the Join Key

Before joining, `count()` and `filter()` were used to check that each participant (`SEQN`) appears only once per cycle (`wave_id`) in each combined dataset. A join on a key that is not unique would multiply rows unexpectedly.

Table 6: Participants appearing more than once per cycle

dataset	duplicated_keys
<code>pressure_all_cycles</code>	0
<code>demographic_all_cycles</code>	0

Join Datasets

A `full_join()` was used to keep every record from both combined tables, including any that have no match in the other table. Records are matched on the participant identifier `SEQN`; the cycle indicator `wave_id` is included in the join key as well, so that a participant is only ever matched within the same survey cycle. `anti_join()` was then used to count the records on each side that found no match.

Table 7: Merge summary

blood_pressure_rows	demographic_rows	merged_rows	merged_columns	unmatched_pres	unmatched_demo
28061	29400	29400	19	0	1339

Table 8: Snippet of the merged dataset (demo_pres_merged)

SEQN	wave_id	RIAGENDR	RIDAGEYR	BPXSY1	BPXD11
73557	2013-2014	1	69	122	72
73558	2013-2014	1	54	156	62
73559	2013-2014	1	72	140	90
73560	2013-2014	1	9	108	38
73561	2013-2014	2	73	136	86
73562	2013-2014	1	56	160	84

The merged dataset contains 29,400 rows and 19 columns. It has the same number of rows as the demographic dataset (29,400) because `full_join()` keeps all records from both tables and every blood pressure record found a demographic match: there were 0 unmatched blood pressure records, while 1,339 demographic records had no matching blood pressure record (participants who were interviewed but did not receive a blood pressure exam). The duplicate check found 0 duplicated join keys, so the join did not create extra rows. The merged dataset has 19 columns because the two tables share `SEQN` and `wave_id`, which appear only once after the join.

4 Data Cleaning and Variable Recoding

This section presents a structured cleaning pipeline to prepare the merged data for analysis: reviewing the data, standardizing column names, recoding variables into factors, creating age groups, and handling duplicates and invalid blood pressure readings.

Review the Merged Data

The `skim_without_charts()` function from `skimr` was used to review `demo_pres_merged`. A missing-value summary was then created with `summarise(across(...))` and `pivot_longer()` to show how many values are missing in each column and what percentage of the rows that represents.

Table 9: Data summary

Name	demo_pres_merged
Number of rows	29400
Number of columns	19
Column type frequency:	
character	1
numeric	18
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
wave_id	0	1	9	9	0	3	0

Variable type: numeric

Table 10: Missing values per column before cleaning

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
SEQN	0	1.00	88256.50	8487.19	73557	80906.75	88256.5	95606.25	102956
PEASCCT1	28400	0.03	20.64	30.89	1	3.00	3.0	56.00	122
BPXSY1	8781	0.70	119.94	18.91	66	106.00	116.0	130.00	236
BPXDI1	8781	0.70	66.55	15.20	0	58.00	68.0	76.00	136
BPXSY2	8115	0.72	119.99	19.11	66	106.00	116.0	130.00	238
BPXDI2	8115	0.72	66.46	15.46	0	58.00	68.0	76.00	144
BPXSY3	8178	0.72	119.67	18.90	62	106.00	116.0	130.00	238
BPXDI3	8178	0.72	66.36	15.75	0	58.00	68.0	76.00	140
BPXSY4	28005	0.05	128.18	23.92	74	110.00	126.0	144.00	234
BPXDI4	28005	0.05	71.30	17.27	0	62.00	72.0	82.00	132
RIAGENDR	0	1.00	1.51	0.50	1	1.00	2.0	2.00	2
RIDAGEYR	0	1.00	32.52	24.91	0	10.00	28.0	54.00	80
RIDRETH1	0	1.00	3.11	1.29	1	2.00	3.0	4.00	5
RIDRETH3	0	1.00	3.33	1.67	1	2.00	3.0	4.00	7
DMDEDUC2	12343	0.58	3.49	1.26	1	3.00	4.0	4.00	9
DMDMARTL	12343	0.58	2.64	2.79	1	1.00	1.0	5.00	99
DMDFMSIZ	0	1.00	3.70	1.78	1	2.00	4.0	5.00	7
INDFMIN2	926	0.97	11.28	15.34	1	5.00	8.0	14.00	99

variable	n_missing	pct_missing
SEQN	0	0.0
PEASCCT1	28400	96.6
BPXSY1	8781	29.9
BPXDI1	8781	29.9
BPXSY2	8115	27.6
BPXDI2	8115	27.6
BPXSY3	8178	27.8
BPXDI3	8178	27.8
BPXSY4	28005	95.3
BPXDI4	28005	95.3
wave_id	0	0.0
RIAGENDR	0	0.0
RIDAGEYR	0	0.0
RIDRETH1	0	0.0
RIDRETH3	0	0.0
DMDEDUC2	12343	42.0
DMDMARTL	12343	42.0
DMDFMSIZ	0	0.0
INDFMIN2	926	3.1

The merged dataset contains 29,400 rows and 19 columns, made up of 18 numeric variables and 1 character variable (`wave_id`).

Missing values varied widely across variables. The blood pressure comment variable (PEASCCT1) had 28,400 missing values (96.6%), which is expected because it is only filled when the technician noted a problem. The fourth systolic and diastolic readings (BPXSY4 and BPXD14) had 28,005 and 28,005 missing values (95.3% and 95.3%), since a fourth reading was rarely needed. The first three blood pressure readings had between 27.6% and 29.9% missing.

Among the demographic variables, DMEDEDUC2 had 12,343 missing values (42%) and DMDMARTL had 12,343 (42%), while INDFMIN2 had 926 (3.1%). Education and marital status are not recorded for younger participants, which explains most of their missing values. This review gave an initial assessment of data completeness and identified the variables that need attention during cleaning and analysis.

Standardize Column Names

`clean_names()` from `janitor` converted all column names to lower case with underscores, and `rename()` then gave each column a clear, descriptive name (for example, `seqn` became `participant_id` and `bpxsy1` became `systolic_bp_1`).

```
## [1] "participant_id"      "blood_pressure_comment" "systolic_bp_1"
## [4] "diastolic_bp_1"      "systolic_bp_2"         "diastolic_bp_2"
## [7] "systolic_bp_3"       "diastolic_bp_3"         "systolic_bp_4"
## [10] "diastolic_bp_4"      "survey_cycle"          "sex"
## [13] "age_years"           "race_1"                "race_3"
## [16] "education"           "marital_status"        "family_size"
## [19] "family_income"
```

Recode Variables and Create Age Groups

The categorical demographic variables (gender, ethnicity, education, marital status, family income, and family size) were converted from numeric codes to factors using `factor()` with the categories in the NHANES data dictionary. Any code that is not listed in `levels` (such as “refused” or “don’t know”) becomes `NA` automatically. Age was grouped into five categories (0–19, 20–39, 40–59, 60–79, and 80+) with `cut()`, where `right = FALSE` makes each group start at its lower bound. The identifier and survey cycle were also converted to character and factor types respectively.

Table 11: Participants by gender

gender	n
Male	14452
Female	14948

Table 12: Participants by age group

age_group	n
0–19	12343
20–39	5594
40–59	5571
60–79	4737
80+	1155

Table 13: Participants by education level

education_level	n
Less than 9th grade	1622
9th–11th grade	2105
High school graduate/GED	3864
Some college/AA degree	5240
College graduate or above	4201
NA	12368

Table 14: Snippet of the recoded variables

participant_id	survey_cycle	gender	age_years	age_group	ethnicity_3	marital_status
73557	2013-2014	Male	69	60–79	Non-Hispanic Black	Divorced
73558	2013-2014	Male	54	40–59	Non-Hispanic White	Married
73559	2013-2014	Male	72	60–79	Non-Hispanic White	Married
73560	2013-2014	Male	9	0–19	Non-Hispanic White	NA
73561	2013-2014	Female	73	60–79	Non-Hispanic White	Married
73562	2013-2014	Male	56	40–59	Mexican American	Separated

The frequency tables confirm that the numeric codes were converted into the intended labelled categories; the NA rows in the education table are the participants for whom education was not recorded.

Removing Duplicates and Invalid Pressure Readings

Fully duplicated rows were removed with `distinct()`, and an additional check counted whether any participant appears more than once within a survey cycle.

The `blood_pressure_comment` variable (PEASCCT1) is filled when the technician notes a problem with the measurement. Any row with a comment is therefore treated as an invalid reading, and its eight blood pressure measurements are set to missing using `mutate(across(...))` with `if_else()`. Counts before and after confirm that the step worked as intended.

0 fully duplicated rows were removed, and 0 participants appear more than once within a survey cycle. ‘

1,000 rows carried a technician comment, so their blood pressure readings were set to missing. This reduced the number of recorded readings from 129,042 to 129,014, and 0 readings remain on the commented rows. Although 1,000 rows carried a technician comment, only 28 readings were removed, which shows that most commented rows already had missing blood pressure values.

The cleaned dataset was saved as `demo_pres_clean_cols.rds` in the `output` folder.

5 Filtering and Reshaping Dataset

This section filtered the cleaned data and reorganized it into a format suitable for analysis and for comparison across readings.

Filtering participants 21 Years and Older and reshaping Pressure Readings

`filter()` kept only adults aged 21 years or older with a known age, and `select()` retained only the variables required for the analysis, producing the wide-format dataset `data_analyze`.

Reshaping the Pressure Readings

`pivot_longer()` reshaped the four systolic and four diastolic reading columns from wide to long format. The `names_pattern` argument splits each column name into its measure (`systolic_bp` or `diastolic_bp`, which become column names via `value`) and the reading number (1 to 4), so each row in the long dataset is one reading for one participant. `mutate()` then converted `reading_number` to an integer.

Table 15: Snippet of the wide-format data (`data_analyze`)

age_years	participant_id	gender	systolic_bp_1	systolic_bp_2	diastolic_bp_1	diastolic_bp_2	systolic_bp_3	diastolic_bp_4
69	73557	Male	122	114	72	76	NA	NA
54	73558	Male	156	160	62	80	NA	NA
72	73559	Male	140	140	90	76	NA	NA
73	73561	Female	136	134	86	88	NA	NA

Table 16: Snippet of the long-format data (`data_analyze_long`)

age_years	participant_id	gender	reading_number	systolic_bp	diastolic_bp
69	73557	Male	1	122	72
69	73557	Male	2	114	76
69	73557	Male	3	102	74
69	73557	Male	4	NA	NA
54	73558	Male	1	156	62
54	73558	Male	2	160	80
54	73558	Male	3	156	42
54	73558	Male	4	NA	NA

Table 17: Blood pressure readings by reading number

reading_number	n_rows	valid_systolic	mean_systolic	mean_diastolic
1	16787	14679	125.4	70.7
2	16787	15214	125.4	70.6
3	16787	15189	124.9	70.5
4	16787	1060	134.6	74.9

The filtered wide-format dataset (`data_analyze`) contains 16,787 rows and 11 columns. After pivoting, `data_analyze_long` contains 67,148 rows and 6 columns.

The long dataset has four times as many rows because each participant's four systolic and diastolic reading pairs became four separate observations, while the number of columns fell from 11 to 6. The reading-number summary shows how many valid readings are available for each reading, which is useful when deciding how to handle the later readings that are often missing.

6 Conclusion

This report presented a data preparation process using NHANES demographic and blood pressure data from the 2013–2014, 2015–2016, and 2017–2018 survey cycles. The datasets were combined, reviewed, and cleaned by standardizing column names, recoding variables, creating age groups, checking for duplicates, and setting blood pressure readings that carried a technician comment to missing.

The data were then filtered to participants aged 21 years and older and reshaped from wide to long format. The filtered wide dataset contained 16,787 rows and 11 columns, while the long-format dataset contained 67,148 rows and 6 columns. These steps organized the data into a consistent, analysis-ready structure for examining demographic characteristics and blood pressure measurements.